

a. Detailed L_2 -Convergence Check

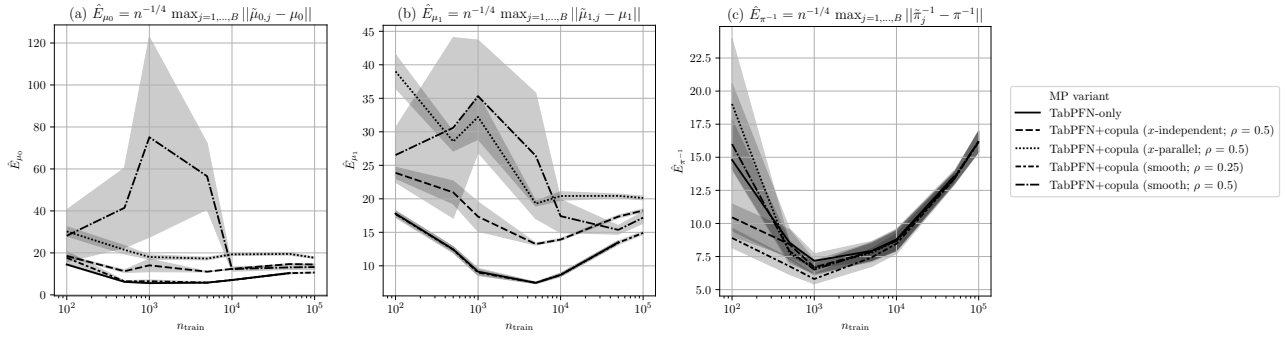


Figure 1. L_2 -convergence check for every nuisance function based on the synthetic data with varying size of the train data, n_{train} (here $d_x = 25$). Reported: mean L_2 error $\hat{E}_\diamond \pm \text{se}$ over 10 runs (lower is better), where $\diamond$ corresponds to different nuisance functions: (a) $\diamond = \mu_0$, (b) $\diamond = \mu_1$, and (c) $\diamond = \pi^{-1}$. **Results.** These results indicate that performance improves with sample size up to a moderate regime, consistent with the L_2 -convergence condition required for the BvM theorem. However, beyond this regime, the concentration of the inverse propensity score posterior deteriorates, which reflects a difficulty of accurately recovering inverse propensity scores for propensities close to zero and one with TabPFN. Together, this confirms that, in practice, the contraction rates are sufficient to yield stable and well-aligned uncertainty estimates, yet only for medium-sized datasets.

b. Memory Usage & Runtime

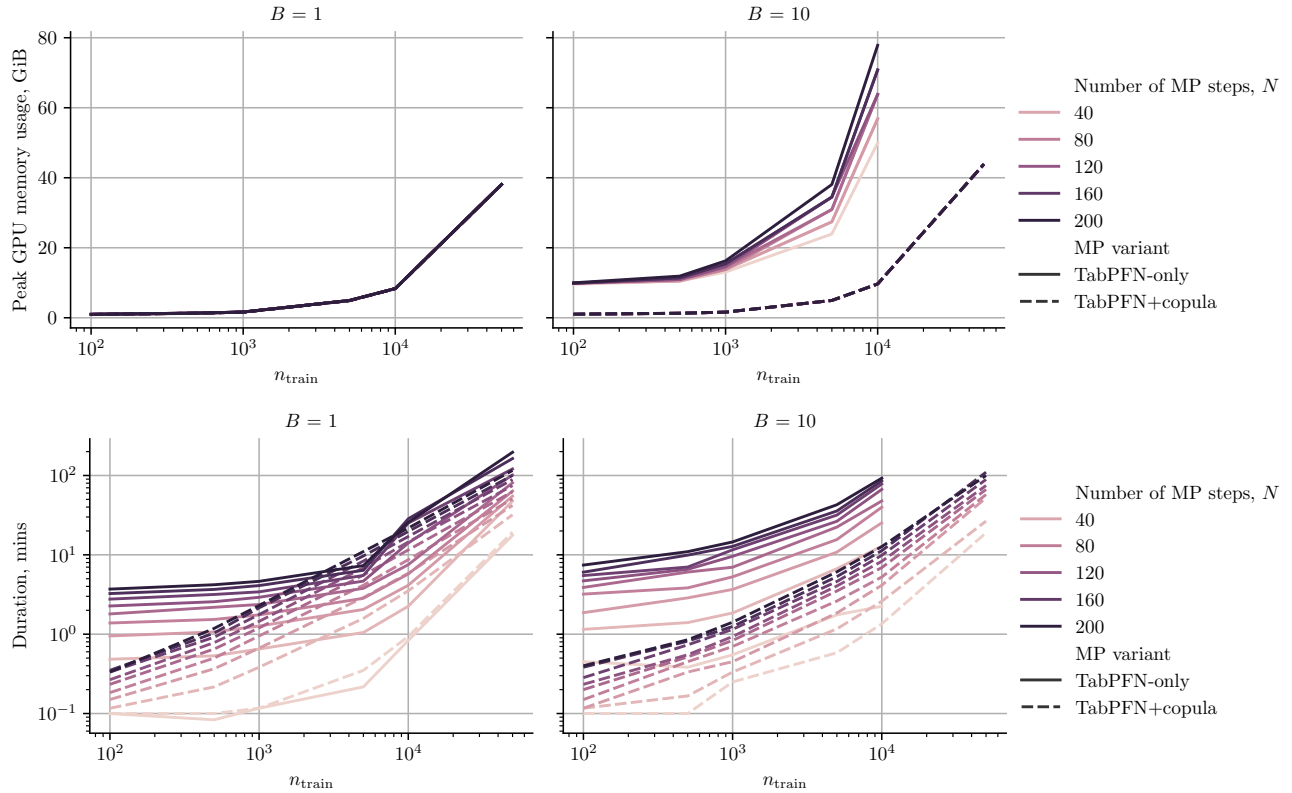


Figure 2. Peak GPU memory usage (top figure, in gigabytes (GiB)) and total runtime (bottom figure, in minutes) of ATE experiments for our *MP-OSPC* with different MP variants and varying number of MP steps based on the synthetic data with varying size of the train data, n_{train} (here $d_x = 25$). Experiments are carried out on AMD EPYC 9455 48-Core CPU and 1 H200 Nvidia GPU. **Results.** While a TabPFN-only MP is very GPU-memory/runtime intensive for increasing number of posterior draws B (as every draw has to be parallelized and requires N subsequent TabPFN calls each with up to $O((n + N)^2)$ time complexity), the copula-based MPs are more GPU-memory/runtime effective (as they require one initial TabPFN call with $O(n^2)$ time complexity and all the MP inference for B draws can be easily vectorized).

c. MP convergence

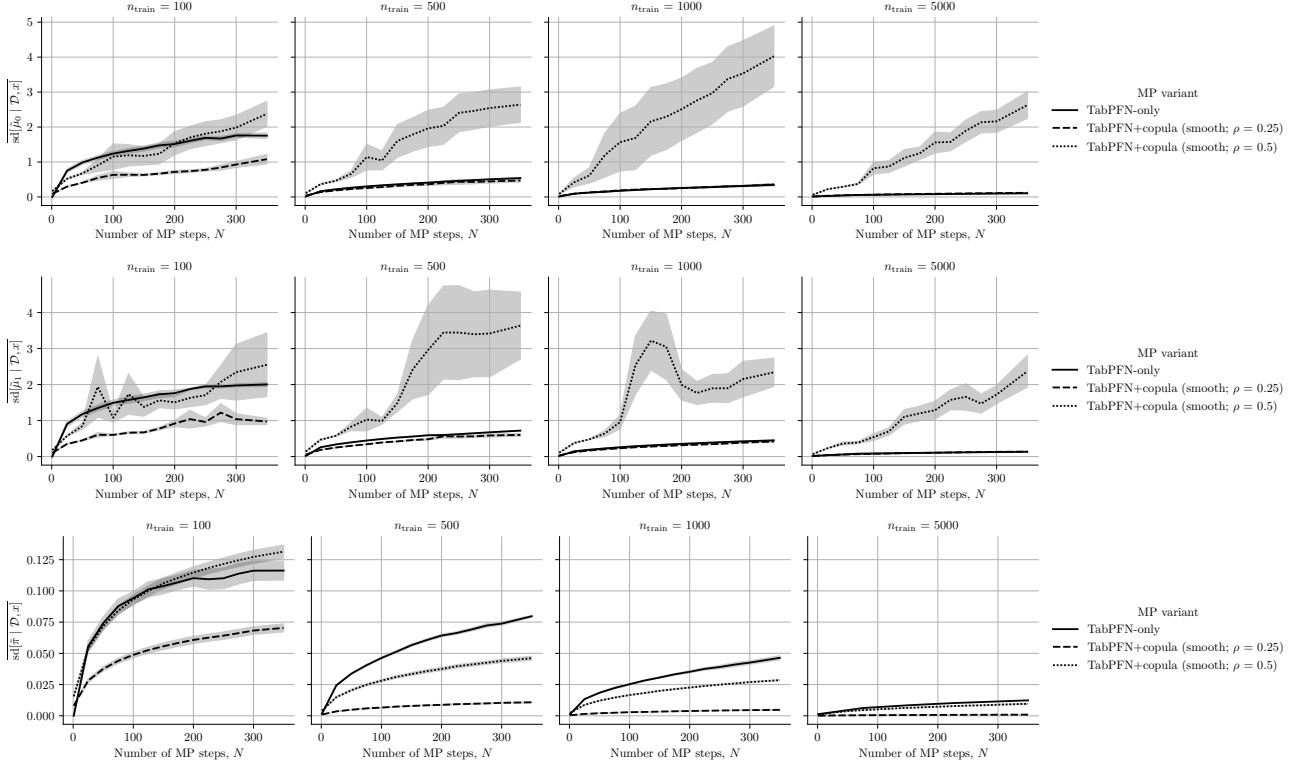


Figure 3. Convergence of different MP variants wrt. increasing number of MP steps N for every nuisance function based on the synthetic data with varying size of the train data, n_{train} (here $d_x = 25$). Reported: mean posterior standard deviation $\text{sd}[\hat{\phi} | \mathcal{D}, x] \pm \text{se}$ over 10 runs, where $\diamond$ corresponds to different nuisance functions: (top row) $\diamond = \mu_0$, (middle row) $\diamond = \mu_1$, and (bottom row) $\diamond = \pi^{-1}$. **Results.** We see that, with few exceptions for large ρ , the posterior standard deviation generally flattens with the growing number of MP steps (e.g., for $N = 300$ with $n = 100$, for $N = 250$ with $n = 500$, etc.)

d. Sensitivity Checks Synthetic Dataset

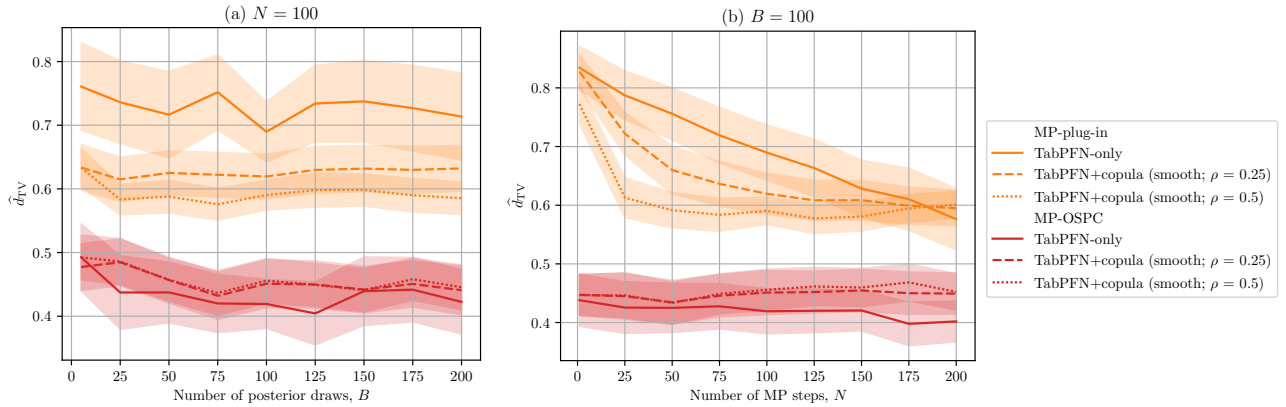


Figure 4. Quality of the asymptotic uncertainty for Bayesian ATE estimators based on the synthetic data with (a) varying size of posterior draws, B , and (b) varying number of MP steps, N (here $n_{\text{train}} = 500$, $d_x = 25$). Reported: mean $\hat{\delta}_{\text{TV}} \pm \text{se}$ over 40 runs (lower is better). **Results.** Our MP-based ATE posteriors converge already when $B = 75$ and $N = 100$. Importantly, the performance of our MP-OSPC remains relatively stable under different hyperparameters highlighting the first-order insensitivity to the misspecification of the nuisance functions posteriors.

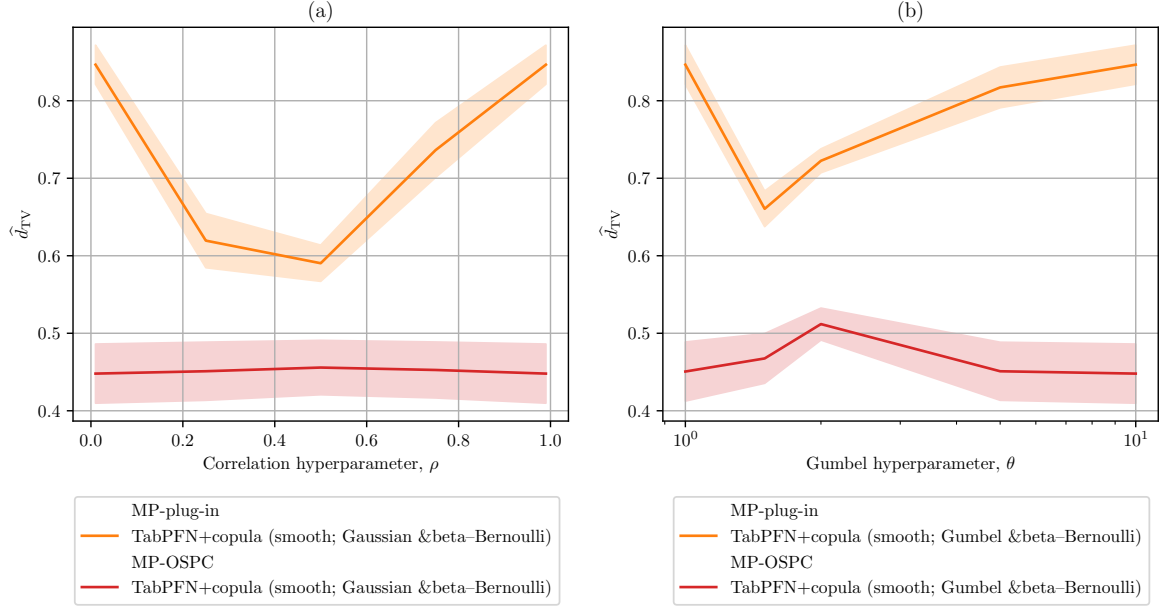


Figure 5. Quality of the asymptotic uncertainty for Bayesian ATE estimators based on the synthetic data with varying copula types, (a) Gaussian and (b) Gumbel, and corresponding hyperparameters, (a) $\rho \in (0, 1)$ and (b) $\theta \geq 1$ ($n_{\text{train}} = 500$, $d_x = 25$, $B = 100$, $N = 100$). Reported: mean $\hat{d}_{TV} \pm \text{se}$ over 40 runs (lower is better). **Results.** While the performance of the MP-plug-ins heavily depends on the choice of copulas and their hyperparameters ρ/θ , the performance of our MP-OSPC remains stable highlighting the first-order insensitivity to the misspecification of the nuisance functions posteriors.

e. Detailed L_2 -Convergence Check for CausalPFN & TabPFN Combination

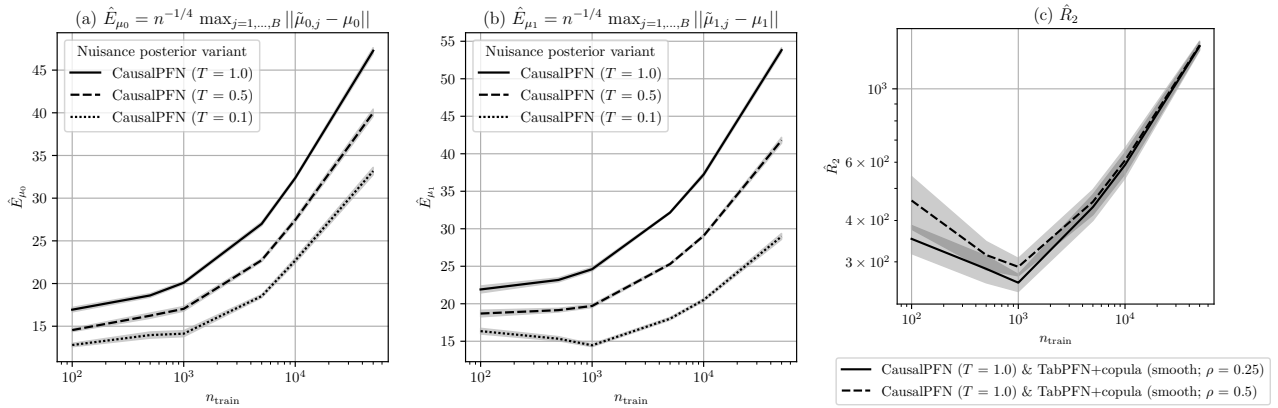


Figure 6. L_2 -convergence check for every nuisance function (a)-(b) and for the second-order remainder $\hat{R}_2$ (c) based on the synthetic data with varying size of the train data, n_{train} (here $d_x = 25$). Reported: mean L_2 error $\hat{E}_{\diamond}/\hat{R}_2 \pm \text{se}$ over 10 runs (lower is better), where $\diamond$ corresponds to different nuisance functions: (a) $\diamond = \mu_0$, (b) $\diamond = \mu_1$ (note that propensity score is not available for CausalPFN). **Results.** These results indicate that performance degrades with increasing sample size for CausalPFN-based functional posteriors μ_a , as seen in (a)-(b). This is not surprising, considering that CausalPFN was trained on synthetic datasets with up to $n \approx 2000$. However, as seen in (c), the second-order remainder $\hat{R}_2$ converges for $n < 1000$. This hints on the double robustness property: the divergence of the CausalPFN-based posteriors for μ_a can be partially compensated with a convergence of TabPFN-based propensity score MP posterior.

f. Synthetic ATE experiments for CausalPFN & TabPFN Combination

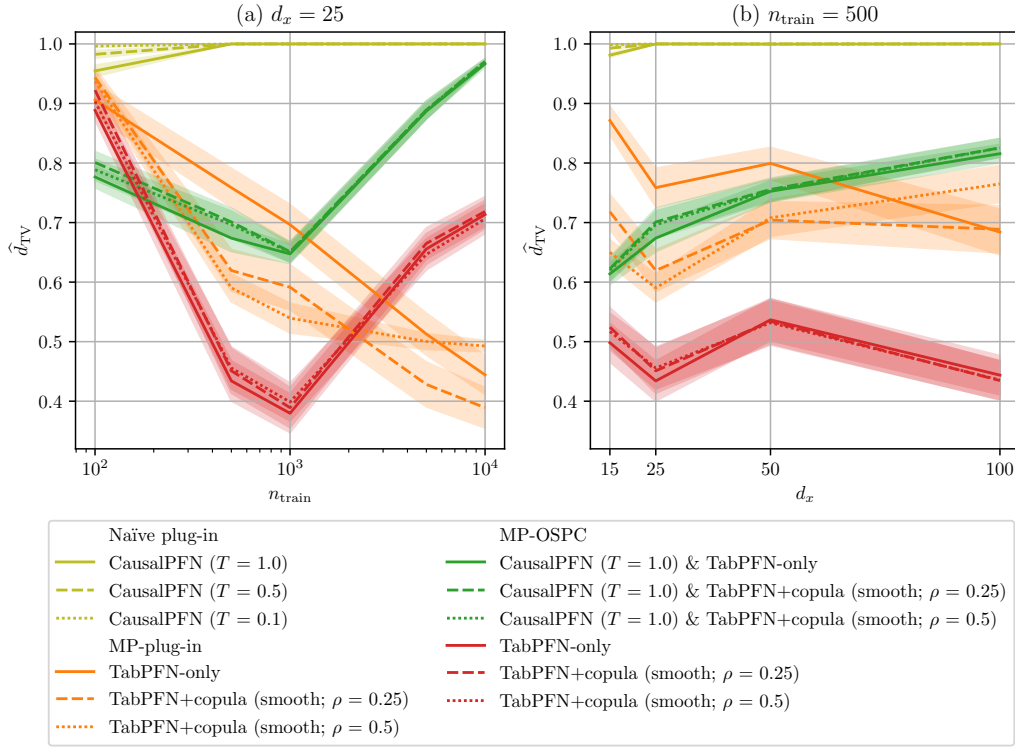


Figure 7. Quality of the asymptotic uncertainty for Bayesian ATE estimators based on the synthetic data with (a) varying size of the train data, n_{train} , and (b) varying dimensionality of covariates, d_x . Reported: mean $\hat{d}_{TV} \pm \text{se}$ over 40 runs (lower is better). **Results.** different variants of our *MP-OSPC* improve over plug-in estimators based on TabPFN and CausalPFN, and often achieve the best asymptotic alignment with the A-IPTW estimator for varying (a) dimensionality of covariates and (b) train data sizes. Notably, the slight drop in performance of *MP-OSPC* for large data sizes (e.g., $n_{\text{train}} > 1000$) is expected and can be attributed to a growing R_2 .

μ_a model	π model	n_{train} ATE estimator	100	500	1000	5000	10000
CausalPFN ($T = 1.0$)	—	Naïve plug-in	0.965	1.000	1.000	1.000	1.000
	TabPFN-only	MP-OSPC	0.685	0.485	0.585	0.890	0.980
	TabPFN+copula (smooth; $\rho = 0.25$)	MP-OSPC	0.735	0.520	0.610	0.865	0.970
	TabPFN+copula (smooth; $\rho = 0.5$)	MP-OSPC	<u>0.695</u>	0.510	0.600	0.865	0.970
CausalPFN ($T = 0.5$)	—	Naïve plug-in	1.000	1.000	1.000	1.000	1.000
CausalPFN ($T = 0.1$)	—	Naïve plug-in	1.000	1.000	1.000	1.000	1.000
TabPFN-only	—	MP-plug-in	0.875	0.855	0.755	0.560	<u>0.305</u>
	TabPFN-only	MP-OSPC	0.865	0.340	0.330	0.820	0.870
TabPFN+copula (smooth; $\rho = 0.25$)	—	MP-plug-in	0.920	0.680	0.690	<u>0.455</u>	0.255
	TabPFN+copula (smooth; $\rho = 0.25$)	MP-OSPC	0.895	0.355	0.370	0.810	0.845
TabPFN+copula (smooth; $\rho = 0.5$)	—	MP-plug-in	0.920	0.590	0.550	0.390	0.335
	TabPFN+copula (smooth; $\rho = 0.5$)	MP-OSPC	0.860	<u>0.350</u>	<u>0.340</u>	0.825	0.830

Lower = better (best in **bold**, second best underlined)

Table 1. Quality of the (ii) finite-sample uncertainty for Bayesian ATE estimators based on the synthetic data with (a) varying size of the train data, n_{train} . Reported: mean $\hat{d}_{KS}$ evaluated with 40 runs, grouped wrt. the underlying PFN. Here: $d_x = 25$. **Results.** Our *MP-OSPC* always improves over naïve or MP-based plug-ins, as long as the data size is $n_{\text{train}} \leq 5000$ (which is expected as TabPFN struggles with consistent estimation of both nuisance functions for large n).